

Replication Plan

Supervised Extraction of Near-Complete Genomes from Metagenomic Samples: A New Service in PATRIC

OSTI ID: 2469515

Auto-generated Replication Blueprint

April 8, 2026

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Paper Overview

This paper presents a supervised metagenome binning pipeline integrated into the PATRIC bioinformatics platform. The workflow takes raw metagenomic sequencing reads, assembles them into contigs using metaSPAdes, bins contigs into metagenome-assembled genomes (MAGs) using a reference-guided (supervised) approach, annotates bins using RASTtk, and evaluates genome quality using EvalG (completeness/contamination via universal marker genes) and EvalCon (annotation consistency). The pipeline is validated on synthetic communities and real environmental metagenomes.

Detailed Workflow

Phase 1: Data Acquisition

1.1 Download metagenomic sequencing reads from NCBI SRA.

- Specific SRA accession numbers referenced in the paper.
- Both synthetic community (mock) and real environmental samples.

1.2 Quality control reads.

- Adapter trimming (Trimmomatic or fastp).
- Quality filtering ($Q \geq 20$).
- Remove host contamination if applicable.

Phase 2: Assembly

2.1 Assemble reads with metaSPAdes.

- `metaspades.py -1 reads_1.fq -2 reads_2.fq -o assembly_output`
- Multiple k -mer sizes (default: 21, 33, 55, 77).

2.2 Assess assembly quality (QUAST or basic N50/contig count statistics).

Phase 3: Supervised Binning

3.1 Map reads back to contigs (BWA-MEM or Bowtie2) to compute coverage.

3.2 Taxonomic classification of contigs.

- BLAST or Kraken2 against a reference database.
- Assign contigs to taxonomic bins based on classification + coverage + composition.

3.3 Supervised binning using SEEDtk tools.

- Reference-guided assignment using PATRIC genome database.
- Iterative refinement of bins.

Phase 4: Annotation and Quality Assessment

4.1 Annotate each bin with RASTtk.

- Gene calling, functional annotation, subsystem classification.

4.2 Evaluate genome quality.

- **EvalG**: completeness and contamination via universal single-copy marker genes.
- **EvalCon**: consistency of annotation with expected genome features.
- Apply MIMAG quality standards (high: > 90% complete, < 5% contamination).

Phase 5: Validation

5.1 Compare MAG quality against known reference genomes (for synthetic communities).

5.2 Compare binning performance against MetaBAT2, MaxBin2, CONCOCT.

5.3 Reproduce quality summary tables from the paper.

Datasets

Dataset / Source	Type	Location / Notes
NCBI SRA reads	Metagenomic reads	https://www.ncbi.nlm.nih.gov/sra Specific accessions from paper
CAMI synthetic community	Benchmark	https://data.cami-challenge.org/
PATRIC genome database	Reference genomes	https://www.bv-brc.org/ (formerly PATRIC)
SEEDtk tools and data	Pipeline code	https://github.com/SEEDtk
Supplementary tables	Quality metrics	Accompanying the paper

Models, Tools, and Software

Tool / Library	Version	Purpose / URL
metaSPAdes	≥ 3.15	Metagenomic assembly https://github.com/ablab/spades
RASTtk	latest	Genome annotation https://rast.nmpdr.org/
PATRIC / BV-BRC	latest	Integrated bioinformatics platform https://www.bv-brc.org/
SEEDtk	latest	Supervised binning tools https://github.com/SEEDtk
BWA-MEM	$\geq 0.7.17$	Read mapping

Tool / Library	Version	Purpose / URL
SAMtools	≥ 1.15	BAM/SAM manipulation
CheckM / CheckM2	latest	Alternative quality assessment https://github.com/Ecogenomics/CheckM
MetaBAT2	latest	Alternative binner for comparison https://bitbucket.org/berkeleylab/metabat
Trimmomatic / fastp	latest	Read quality trimming
Python	≥ 3.8	Scripting and analysis

Hardware Requirements

- metaSPAdes assembly: ≥ 64 GB RAM for real metagenomes.
- Multi-core CPU (≥ 16 cores) recommended.
- ≥ 200 GB storage for reads, assemblies, and databases.